

ML-LUCJ v3: The κ "Gauge Wall" Is a Coordinate Problem ## A learnability study of eigenvector-valued targets, and the fix
Follow-up to deck_v2 — gauge study + retargeted pretraining (Aug 2026)

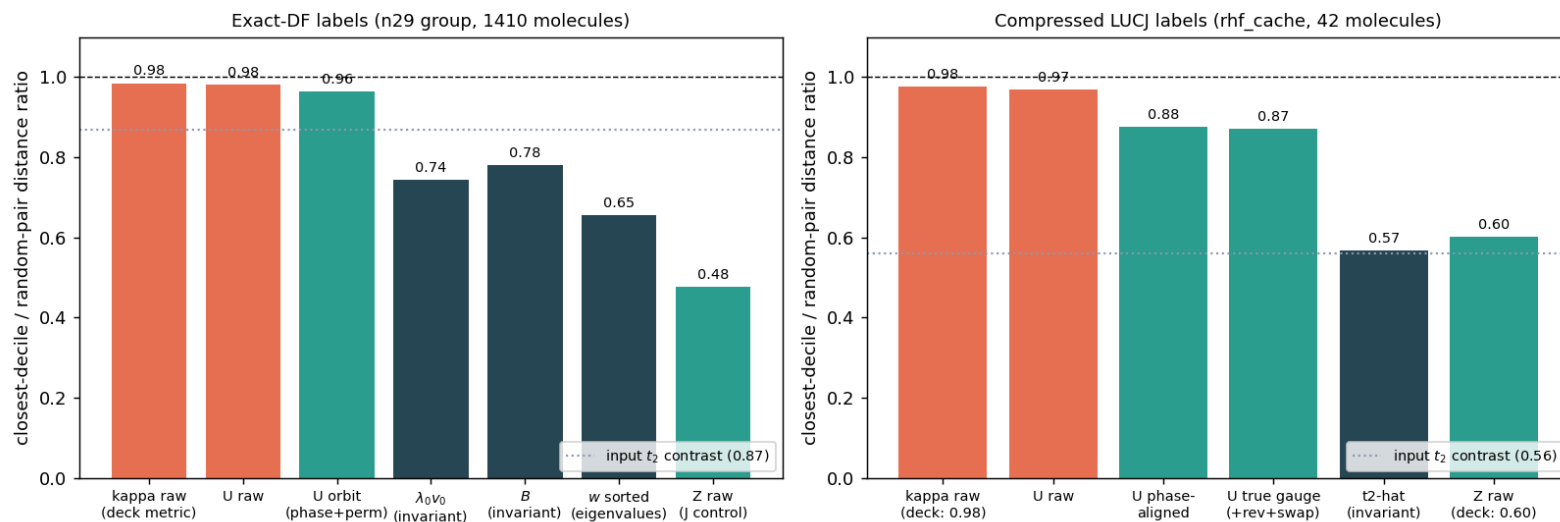
Where v_2 left off — and the question **v_2 conclusion**: J (diagonal Coulomb) is learnable (R^2 0.63–0.80); U/κ (orbital rotation) is "unlearnable" — local-linear $R^2 = -0.16$, nearest-pair ratio 0.98, κ val MSE flat from epoch 1 (0.1028 for 40 epochs). **The challenge**: is an eigenvector-valued objective *truly* unlearnable — in any sense? **Method of this study** (all on our data; code in `gauge\_study/`, full report in `docs/eigenvector\_learnability\_study.md`): 1. Pin down mathematically what the (U, Z) labels *are* 2. Distinguish four meanings of "unlearnable" and test each 3. Re-measure v_2 's smoothness/learnability tests with **gauge-aware distances and metrics** 4. Survey how other fields learn eigenvector-valued outputs 5. **Fix the objective, retrain, compare**

What the labels actually are (verified to float32 on our targets) `generate\_df\_targets.py` (optimize=False) is a *nested eigendecomposition*: $T_{(ia),(jb)} = t_2[i,j,a,b] \rightarrow \text{eigh}$; $T = \sum_k \lambda_k v_k v_k^T$ (sorted by $|\lambda_k|$) $M = \text{mat}(v_0), \text{quad } Q^{\text{pm}} = \frac{1}{2}(1 \mp i)(M^{\text{pm}} M^T)$, $\text{quad } \text{eigh}(Q^{\text{pm}}) = (w^{\text{pm}}, U^{\text{pm}})$ | Identity (50 molecules) | deviation | $|Z_1 - Z_0|$, rep 1 = conj(rep 0) up to gauge | $0 / 1e-18$ | $Z = \text{pm} \lambda_0$, $w w^T$ — **rank-1** | $2e-8$ | invariant content of (U,Z) = $\lambda_0 v_0 v_0^T$ = best rank-1 approx of T | $7e-8$ | zero-weight columns of U ($w_p = 0$, pure gauge) | exactly $n_{\text{occ}} - n_{\text{virt}}$ | **The whole $n_{\text{reps}}=2$ label is deterministic linear algebra on one scalar + one $n_{\text{occ}} \times n_{\text{virt}}$ matrix.** **Hazard found**: same molecule, different LAPACK build → Z reproduces to $1e-9$ but κ changes by rel. distance 1.44 ($\approx$ random). Regenerating targets on another machine silently rewrites every κ label.

Four readings of "k is unlearnable" — only one is true (and it's fixable) | Reading | Verdict | Evidence | | :--|:--|:--| | **V1** No measurable map $t_2 \rightarrow \text{valid}(U, Z)$ exists | **False** — ffsim computes one | trivial | | **V2** No *globally continuous* selection of U exists | **True but benign** — obstruction only at spectral degeneracies (topological; canonicalization provably can't fix it — Dym et al. ICML'24); those directions carry $\approx$ zero Z-weight | theory + gap census | | **V3** The labels *as generated* are not smooth in t_2 | **True — this is what v2 measured**. LAPACK redraws the gauge per input: a label-pipeline artifact, not a task property | exp1/3/5 | | **V4** The *task* — predict any representative with the right invariant content — is unlearnable | **False** — invariant content is as smooth as J and predictable | exp2/4/6 | **v2's evaluation (raw entrywise MSE / $\$R^2$) cannot distinguish V3 from V4.** It measured the gauge draw.

Evidence 1 — Same labels, different coordinates

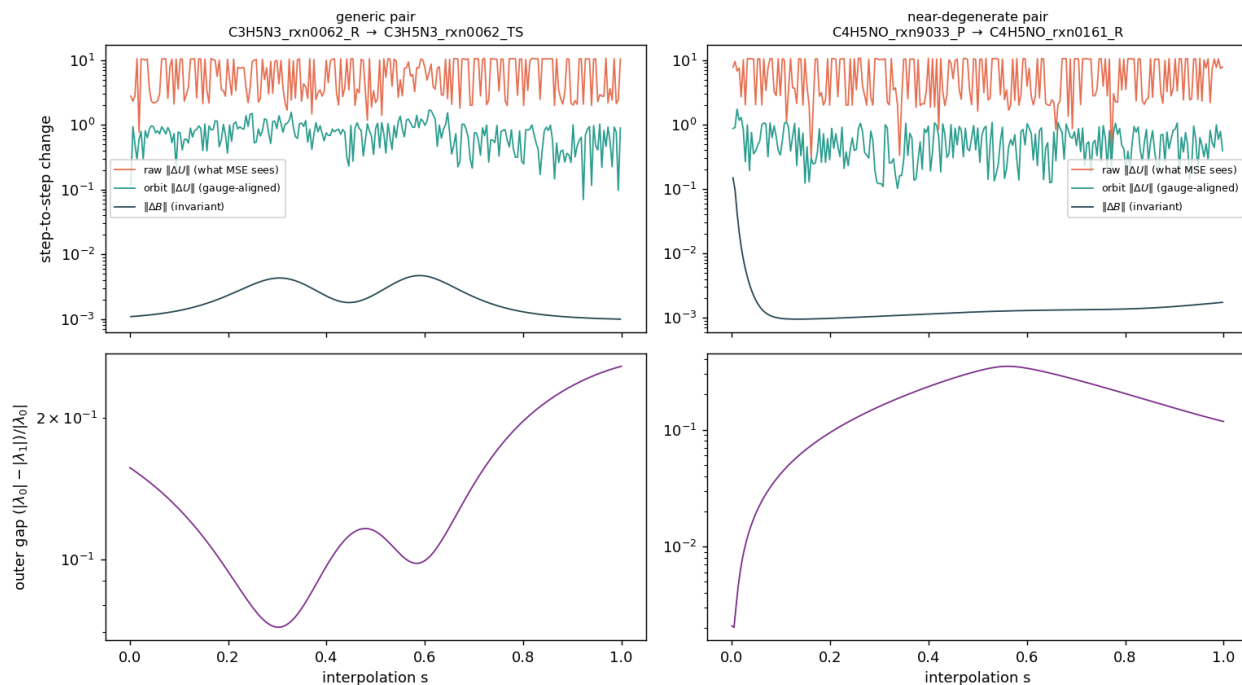
Same molecules, same labels, different coordinates: the invariant content of U is as smooth as the 'learnable' J



On v2's own 42-molecule cache: raw κ ratio **0.976** (deck: 0.98) → phase-aligned **0.876** → the gauge-invariant content of those same labels: **0.568** — smoother than the "learnable" Z (0.601).

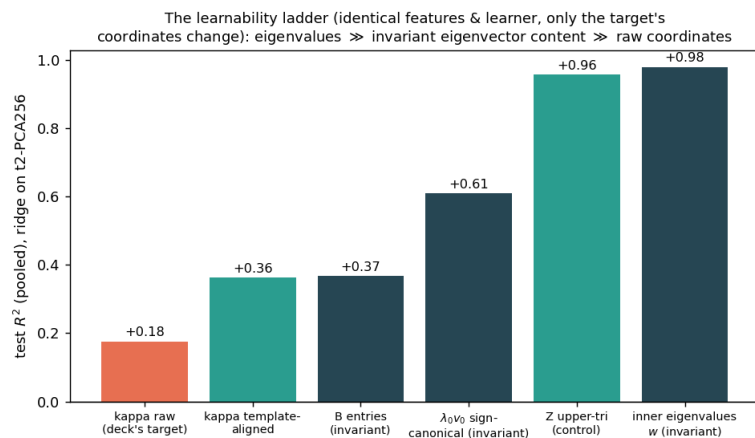
Evidence 2 — Interpolate t2, regenerate labels at every step

Label continuity along a t2 path: raw labels jump everywhere; the orbit/invariant is smooth away from spectral crossings



Raw $\|\Delta U\|$ sits at the ****random-gauge ceiling at every step**** (what κ -MSE supervises). The invariant content moves $\sim 1000\times$ less — and spikes exactly where the outer gap $(|\lambda_0| - |\lambda_1|)/|\lambda_0|$ pinches (Davis-Kahan). Degeneracy census: ****8.4%**** of molecules within 5% of a crossing, 0.6% within 1% — hard cases are ***rare and flaggable by the predicted gap***.

Evidence 3 — The learnability ladder



Identical features (t2-PCA256), identical learner (ridge), 1410 same-shape molecules — only the target's **coordinates** change: | Target | test R^2 | | :-- | :-- | | κ raw (v2's target) | +0.18 | | κ template-aligned | +0.36 | | **$\lambda_0 v_0$ sign-canonical (invariant)** | **+0.61** | | Z upper-tri (control) | +0.96 | | inner eigenvalues w | +0.98 | Metric check: the **same** 1-NN predictor scores 0.77 (vs random) under raw distance but 0.62 under orbit distance. **The metric, not the signal, produced the "unlearnable" verdict.**

What the literature does with eigenvector-valued outputs | Field | Solution | Reference | |:-|:-|:-| | Quantum ML ($H \rightarrow$ orbitals) | ****Never regress eigenvectors****: predict the operator, diagonalize at inference | PhiSNet '21, DeepH '22, QHNet '23 | | ...without labels | self-consistency loss (= our reconstruction loss, mainstream-validated) | Zhang et al. ICML'24 | | Spectral GNNs (eigvecs as inputs) | sign/basis-invariant nets; projectors VV^T | SignNet/BasisNet ICLR'23 | | Canonicalization theory | ****continuous canonicalization provably impossible****; weighted frames | Dym et al. ICML'24 | | Speech / detection | min-over-permutation (Hungarian) losses — orbit losses at scale | PIT '17, DETR '20 | | 3D vision | discontinuous charts (Euler/quaternion) fail regression $\rightarrow$ continuous representations | Zhou et al. CVPR'19 ($\kappa = \log U$ has branch cuts: same mistake) | | VQE warm starts | amortized + generative initializers; predict-then-refine | Qracle '25, Flow-VQE '25 |

The fix: predict the invariant generator, not the gauge

Edge Transformer (unchanged)

pair tokens (B, norb, norb, d)



NEW: v_0 head — scalar on **occ-virt pair tokens** $\rightarrow \hat{m} \in \mathbb{R}^{(n_{\text{occ}} \times n_{\text{virt}})}$

NEW: λ_0 head — masked-mean pool $\rightarrow$ scalar
(J/Z head kept as before; κ heads retired)

▼ inference only

ffsim's own linear algebra (15 lines)

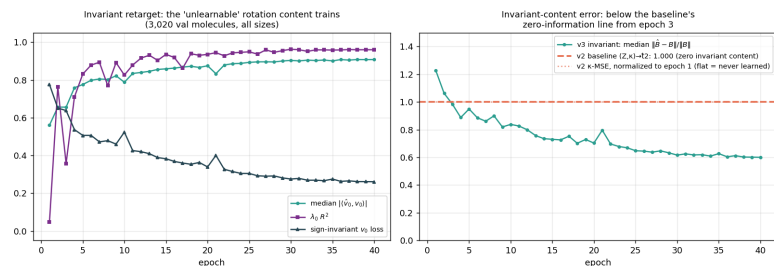
$M = \text{mat}(\hat{v}_0) \rightarrow Q^\pm \rightarrow \text{eigh} \rightarrow (U, Z)$ exactly

****Loss**** (sign is the only residual gauge):
$$\mathcal{L} = \min(\|\hat{m} - v_0\|^2, \|\hat{m} + v_0\|^2) + w_\lambda (\hat{\lambda}_0 - \lambda_0)^2 + \mathcal{L}_Z$$

****Why this is exact, not approximate**** - (λ_0, v_0) is a *sufficient statistic*: reconstruction equals ffsim's output up to gauge (verified 1e-8) - Smooth wherever the outer gap is open; risky 8% flaggable - eigh at inference only $\rightarrow$ no degenerate-eigh backprop - Targets computed from t2 directly (numpy eigh) — ****no ffsim, no gauge in the label pipeline**** ****Bonus** — the $Z=0$ trap dies**: reconstruction in these coordinates is $\sum_k \|\hat{m}_k - T\|^2$ a Burer-Monteiro landscape: $m=0$ is a *strict saddle* (escape curvature $\propto |\lambda_k|$), not v_2 's flat plateau.

New evaluation protocol (replaces κ -MSE / raw R^2) | Metric | What it measures | Status | |:-|:-|:-| | downstream QSCI /
 variational energy vs teacher init | the actual deliverable | future work | | $\|\hat{B} - B\|/\|B\|$, $B = \lambda_0 v_0 v_0^T$ (= $\hat{t}^2$
 error for exact-DF) | invariant content error | **primary** | | sign-aligned cosine $|\langle \hat{v}_0, v_0 \rangle|$ |
 eigenvector direction | logged | | λ_0 MSE / R^2 | factor strength | logged | | accuracy stratified by outer gap | separates model
 failure from task ill-posedness | eval script | | raw entrywise κ MSE / R^2 | the LAPACK gauge draw | **never again** |

Retargeted pretraining: results



Same model, data, split, epochs, batch size as v2's run; **only the objective changed** (lr $5e-4$ vs $1.5e-3$ — the invariant loss is $\sim 10\times$ larger in scale). 40 epochs on $1\times V100$ (scail), ~ 8 min/epoch.

| Metric (3,020 val molecules) | v2 baseline (κ heads) | **v3** (invariant heads) | $||\cdot||^2$ | objective trains at all? | κ MSE **flat from epoch 1** (0.1028) | **v0 loss 1.00 $\rightarrow$ 0.26** | | invariant-content error (median) | **1.000** = zero content¹ | ****0.600**** | | sign-aligned cosine $\langle \hat{v}_0, v_0 \rangle$ (median) | — ($\kappa \rightarrow U$ carries none) | ****0.908**** | | $\lambda_0 R^2$ | — | ****0.960**** | | Z MSE (unchanged head) | $2.6e-5$ | $4.0e-5$ | ¹ the v2 model's κ_{im} head learned $\approx$ the mean $\approx 0 \Rightarrow$ its U is essentially real $\Rightarrow$ reconstructed $\hat{t}_2$ has zero real part: the old pipeline carried **no** invariant rotation information. **Stratified by outer gap** (Davis-Kahan in action): gap > 0.2 (43% of val): B err ****0.433****, cos ****0.953**** . gap 0.05–0.2: 0.700 / 0.874 . gap < 0.05 (15%): 0.978 / 0.731 — errors concentrate exactly where the eigenpair is physically ill-conditioned; flaggable at inference.

Limitations and next steps 1. **$n_{\text{reps}} > 2$** : sufficient statistic becomes the top-K eigenpairs $\rightarrow$ predict the rank-K truncation $\sum_k \lambda_k v_k v_k^T$ (a "denoised t_2 "); split into eigenpairs at inference; subspace losses where λ 's cross. 2. **Compressed / optimize=True targets**: add per-molecule L-BFGS refinement at inference (predict-then-refine — matches how ffsim itself derives them from the exact init). Optimizer multimodality means these should *never* be regressed directly (aligned ratio saturates at 0.87 vs invariant 0.57). 3. **Energy-aligned fine-tuning** (Stage C RL with QSCI reward) is unchanged and compatible. 4. **Near-degenerate molecules (~8%)**: flag by predicted gap; fall back to rank-2 subspace prediction. 5. Label pipeline: pin a gauge + add a cross-BLAS determinism test if κ labels are ever regenerated. **Artifacts**: study `docs/eigenvector\_learnability\_study.md` · experiments `gauge\_study/exp1-6` · figures `gauge\_study/figs/` · this deck's numbers: exp logs committed alongside.